Umjetna Inteligencija

Luka Šiktar

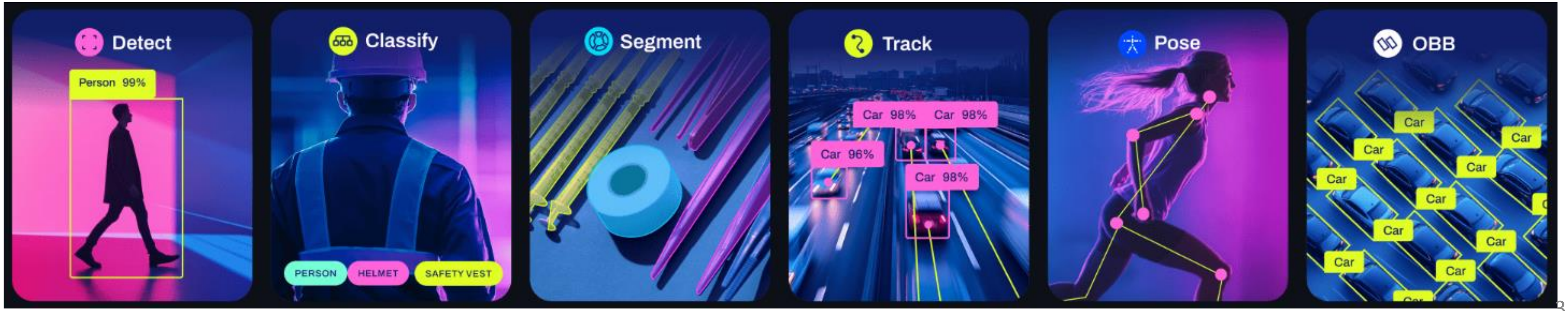
2026 | Regional Center of Excellence for Robotic Technology

Plan Vježbi

- Ultralytics i YOLOv8/YOLOv11 - upoznavanje sa korištenim modelima
- Roboflow – upoznavanje i anotacija setova podataka
- Treniranje modela korištenjem Osobnog računala/Google Colaba

Ultralytics and YOLO

- Ultralytics YOLO Docs - [Ultralytics]
- YOLOv8:
  - State-of-the-art modeli konvolucijskih neuronskih mreža koje su brze, točne i lagane za korištenje
  - Modeli trenirani na [COCO] setu podataka (80 klasa, npr. Auto, bicikl, bus, životinje, objekti, sportska oprema)
  - Detekcija objekata, segmentacija instanci, klasifikacija slika, estimacija skeletona tijela + ostali modovi rada



YOLOv8 Detect

- Object Detection:
  - Zadatak identifikacije klase i lokalizacije objekta na slici ili videu.
  - Ulaz u model: slika
  - Izlaz iz modela: bounding box oko objekta te klasa objekta

YOLOv8 Segment

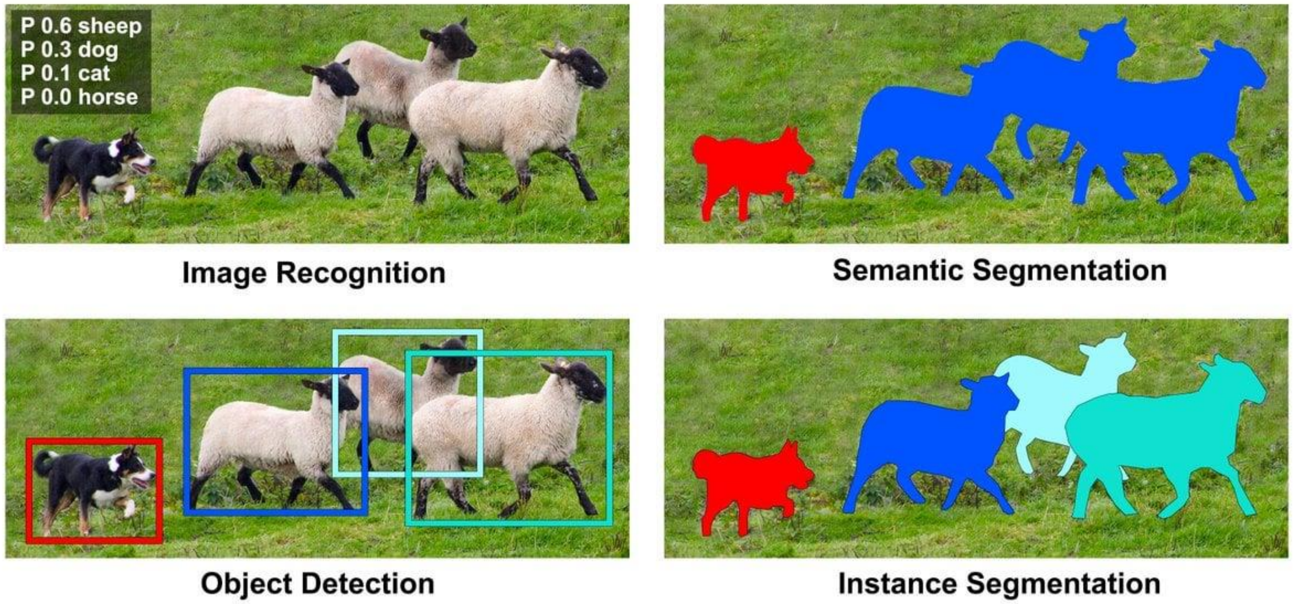
- Instance Segmentation:
    - Zadatak identifikacije klase objekta i segmentacije pojedinog objekta na slici ili videu.
    - Segmentacija – podjela dijelova slike na određene entitete temeljeno na određenim kriterijima (izolacija objekata na slici)
    - Ulaz u model: slika
    - Izlaz iz modela: maska svakog segmentiranog objekta te klasa objekta

YOLOv8 Classify

- Image Classification:
  - Klasifikacije cijele slike u unaprijed definirane klase
  - Ulaz u model: slika
  - Izlaz iz modela: Klasa i vjerojatnost klase na slici

YOLOv8 zadatci

- Primjeri klasifikacije, detekcije objekata i segmentacije:

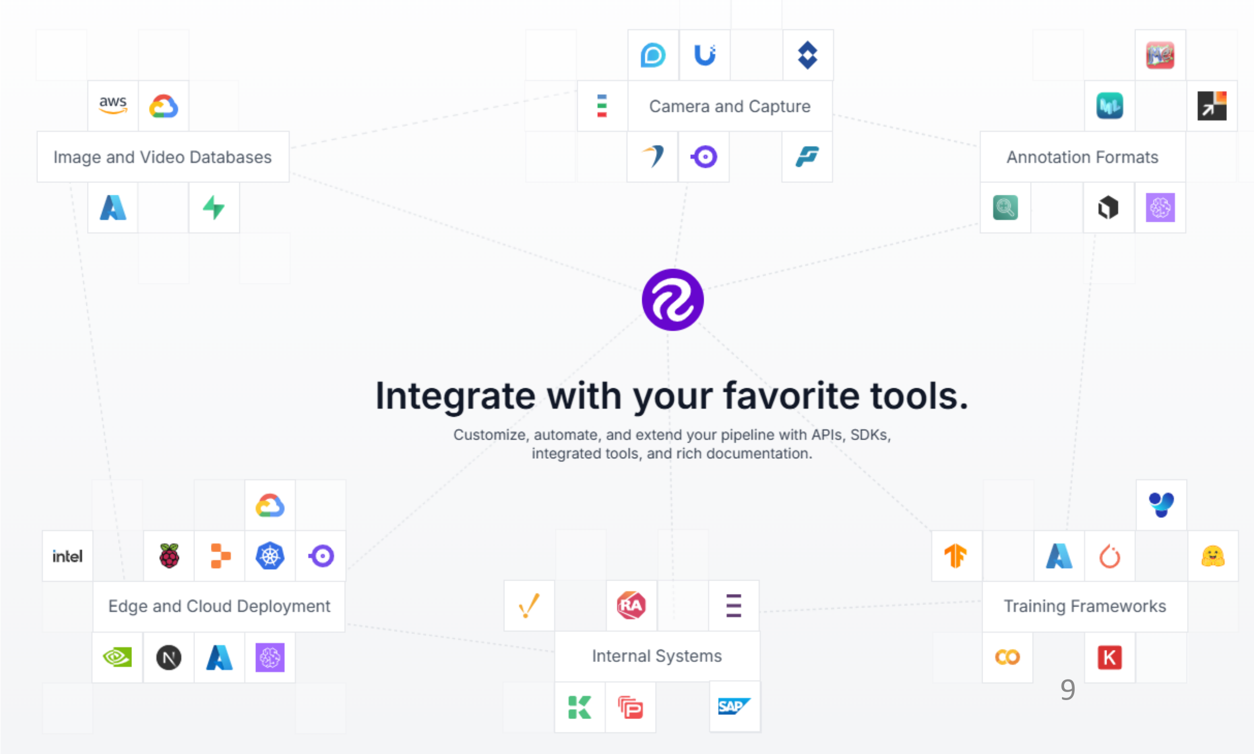


YOLOv8 modovi

- YOLO modeli omogućuju nekoliko modova rada:
    - Trening modela (train mode)
    - Validacija modela (validation mode)
    - Eksploatacija modela (predict mode)
    - Eksploatacija modela, praćenje (track mode)
    - Export mode

Roboflow

- Platforma za stvaranje i implementaciju aplikacija računalnog vida
- Open-source anotatorska platforma za označavanje setova podataka
- Kreirani modeli su open-source



Create a project

Let's create your project.

FMENA > New Public Project

Project Name

E.g., `Dog Breeds` or `Car Models` or `Text Finder`.

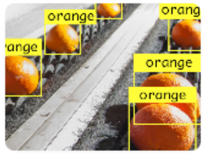
License

CC BY 4.0

Annotation Group

E.g., `dogs` or `cars` or `words`.

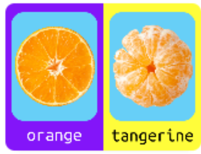
Project Type



Object Detection

Identify objects and their positions with bounding boxes.

Best For # Counting Tracking



Classification

Assign labels to the entire image.

Single-Label Multi-Label

Best For Filtering Content Moderation

Instance Segmentation

Detect multiple objects and their actual shape.

Best For Measurements Odd Shapes

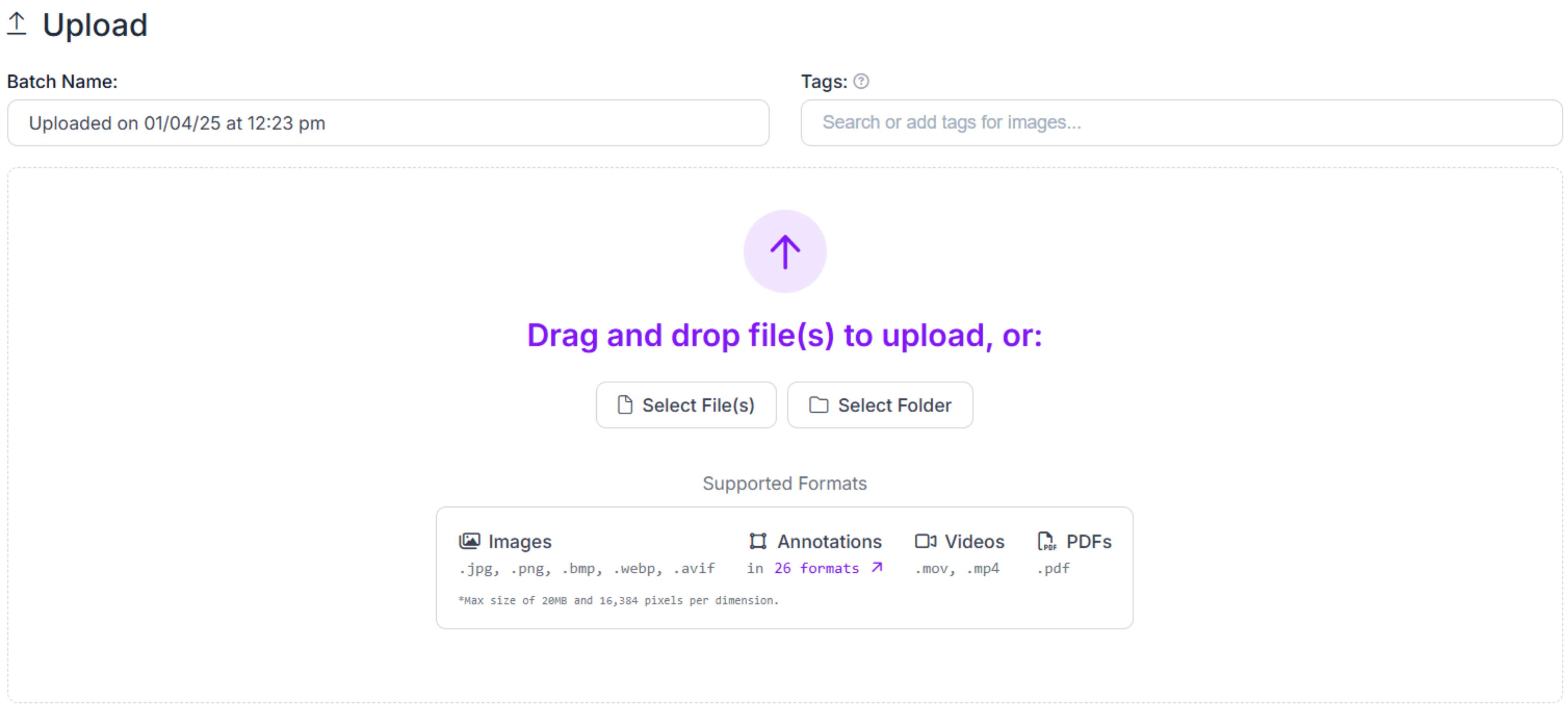
Keypoint Detection

Identify keypoints ("skeletons") on subjects.

Best For Pose Estimation

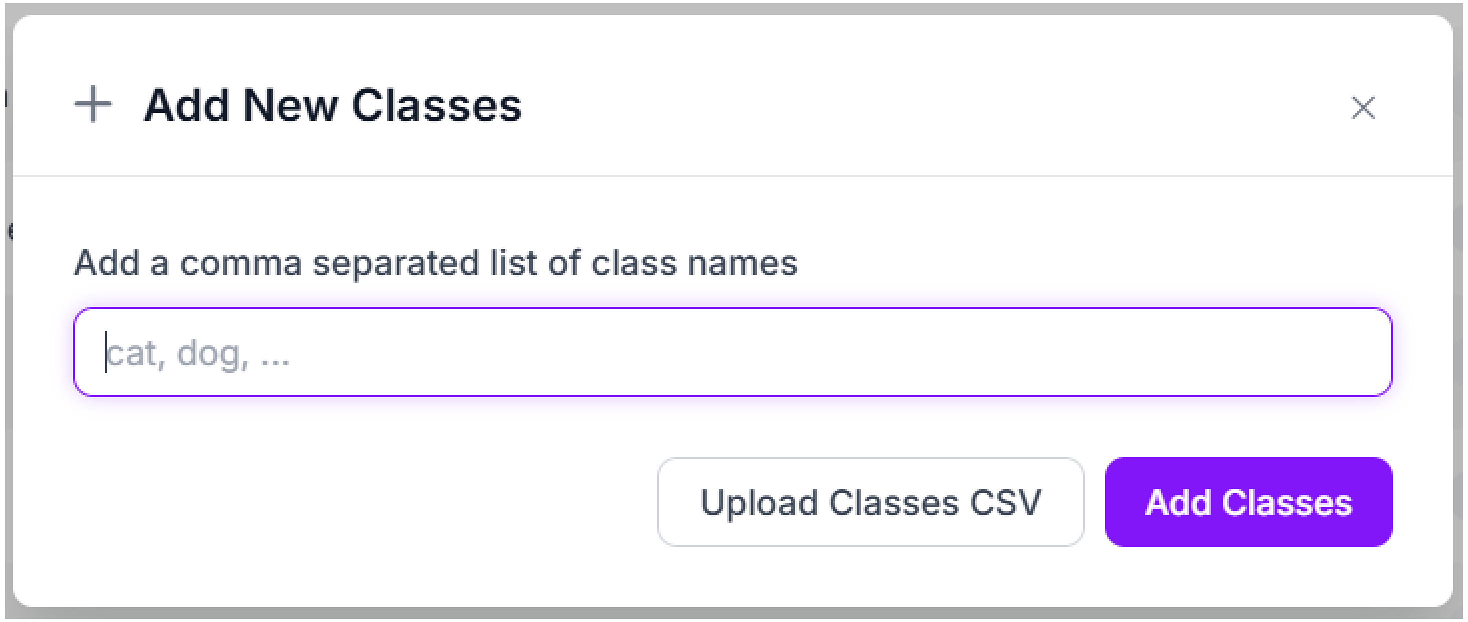
Upload images

Dataset Link: [UmjetnaInteligencija2025Datasetovi]



Upload the classes

Dataset Link: UmjetnaInteligencija2025Datasetovi

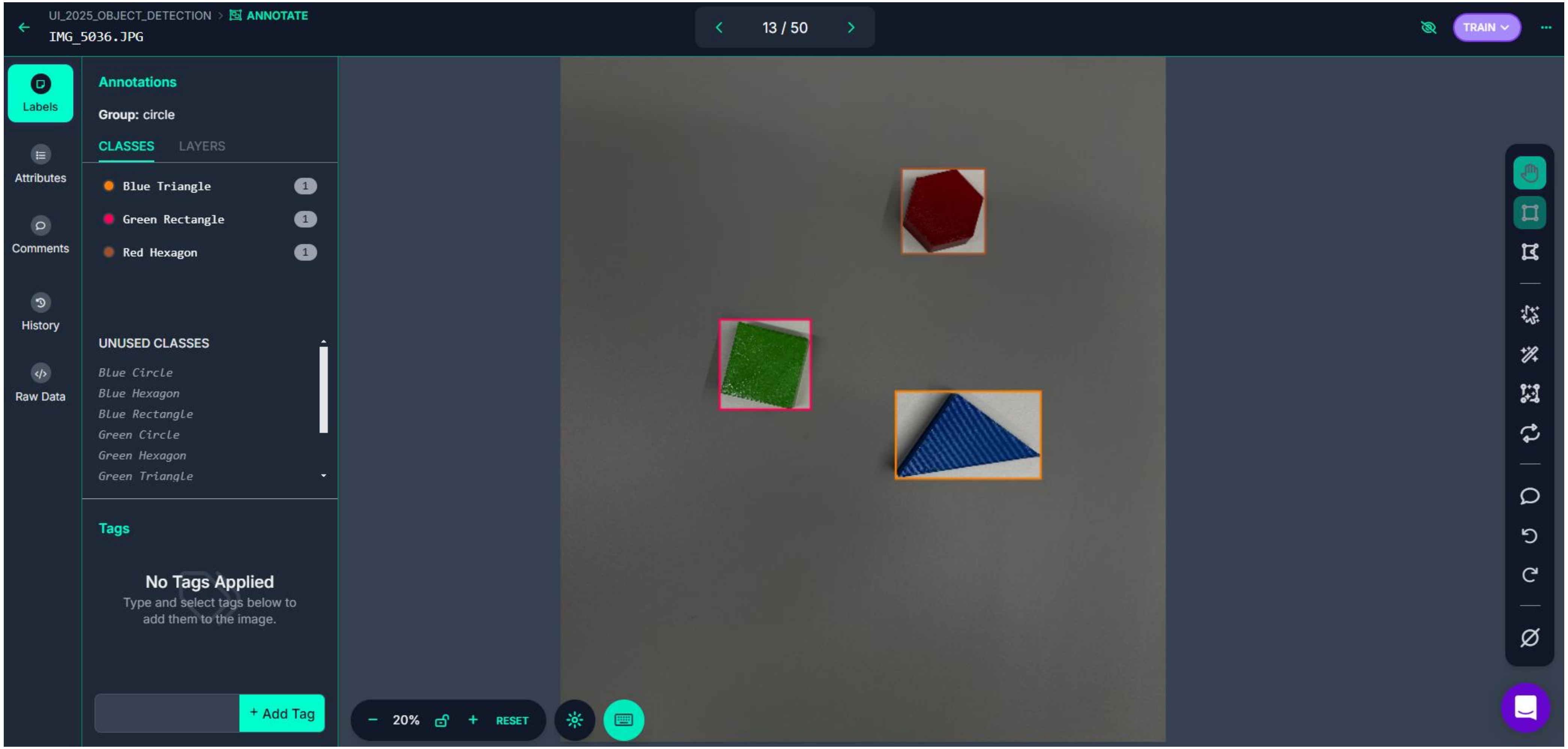


Zadatak 1:Object detection

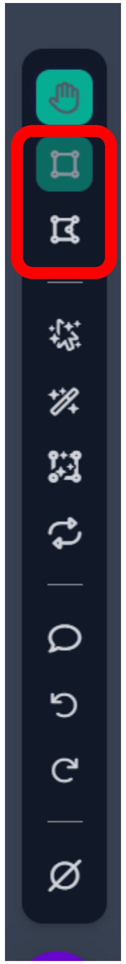
- Cilj: Napraviti set podataka (označiti tražene slike prema danim oznakama) za treniranje YOLOv8 modela
- Potrebno je preuzeti set slika (15 slika) i klase (oznake) prema kojima će se slike označavati
- Uploadati slike i oznake u Roboflow i označiti ih prema pravilima:
    - Za object detection označavanje se koristi pravokutni bounding box (ograničavajuća kućica) ili poligon
    - Cilj je box ili poligon postaviti što bliže rubovima predmeta (Zeleno OK, Crveno NE)
    - Svakom box-u ili poligonu dodati točnu oznaku

Dataset Link: [UmjetnaInteligencija2025Datasetovi]

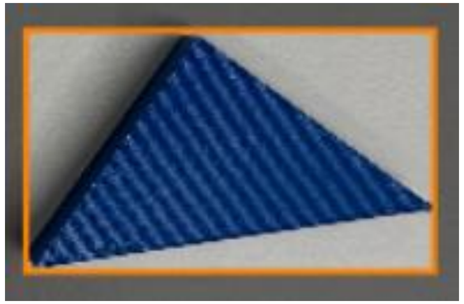
Annotate – Object Detection

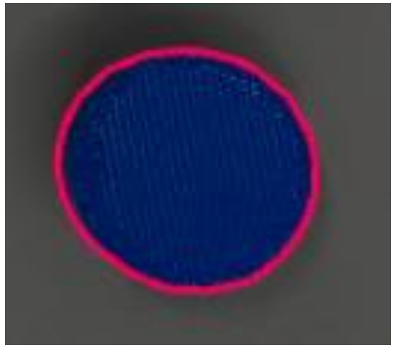


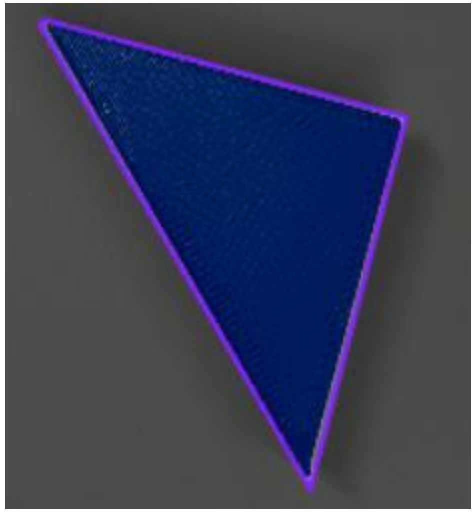
Annotate – Object Detection



- Bounding box tool
- Polygon tool







Create a dataset version

Create New Version

Prepare your images and data for training by compiling them into a version.
Experiment with different configurations to achieve better training results.

Source Images

Images: 127
Classes: 12
Unannotated: 0

Edit

Train/Test Split

Training Set: 89 images
Validation Set: 25 images
Testing Set: 13 images

Create a dataset version

Preprocessing

Auto-Orient: Applied

Resize: Stretch to 640×640

Augmentation

90° Rotate: Clockwise, Counter-Clockwise

Rotation: Between -6° and +6°

Shear: ±10° Horizontal, ±10° Vertical

Noise: Up to 0.1% of pixels

5 **Create**

Review your selections and select a version size to create a moment-in-time snapshot of your dataset with the applied transformations.

Larger versions take longer to train but often result in better model performance. See how this is calculated ↗

Maximum Version Size

305 images (3x)

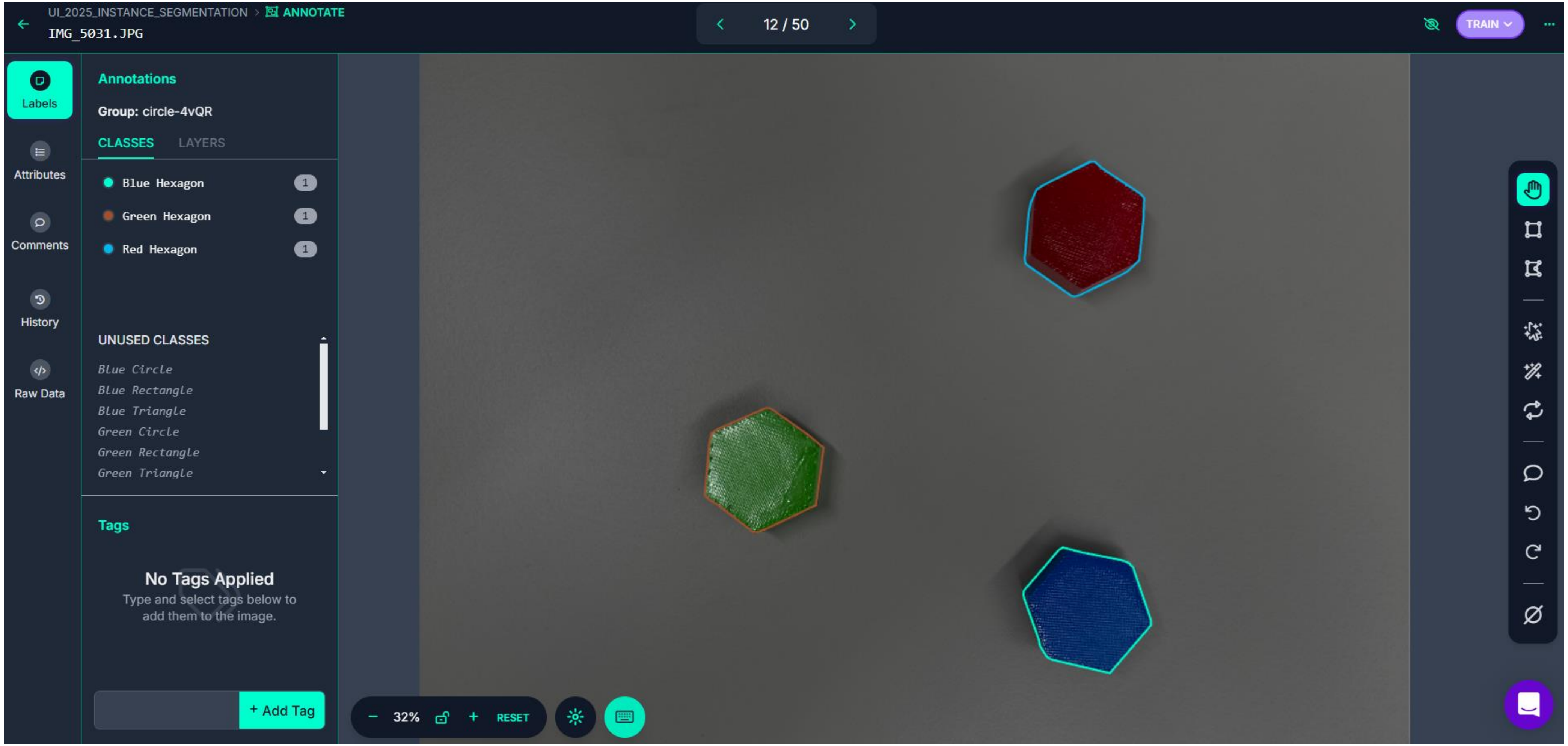
Create

Zadatak 2:Instance segmentation

- Cilj: Napraviti set podataka (označiti tražene slike prema danim oznakama) za treniranje YOLOv8 modela
- Potrebno je preuzeti set slika (15 slika) i klase (oznake) prema kojima će se slike označavati
- Uploadati slike i oznake u Roboflow i označiti ih prema pravilima:
  - Za instance segmentation označavanje se koristi poligon ili AI segmentator SAM2
  - Cilj je poligon postaviti što bliže rubovima predmeta, odnosno modificirati SAM2 segmentaciju kako bi bila što bliže predmetu
  - Svakom poligonu li segmentaciji dodati točnu oznaku

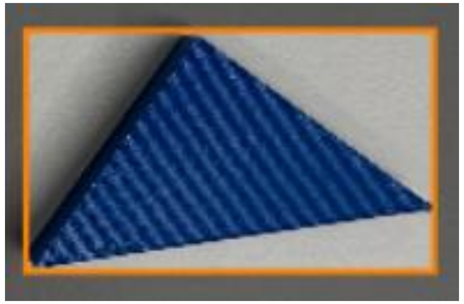
Dataset Link: [UmjetnaInteligencija2025Datasetovi]

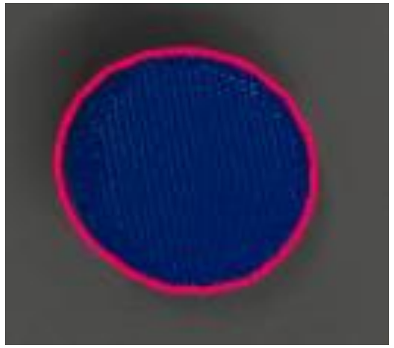
Annotate- Instance Segmentation

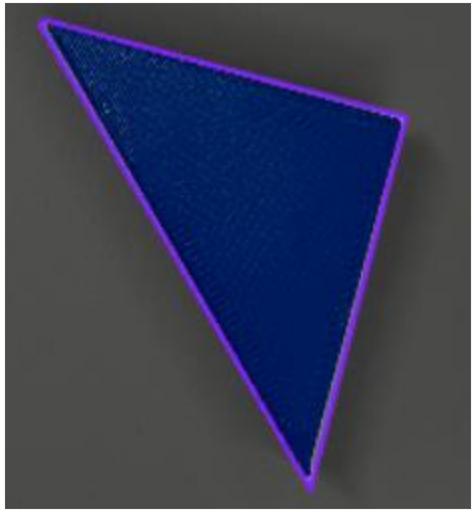


Annotate – Object Detection

- Bounding box tool
- Polygon tool
- **Segementation using AI tools (SAM / SAM2)**





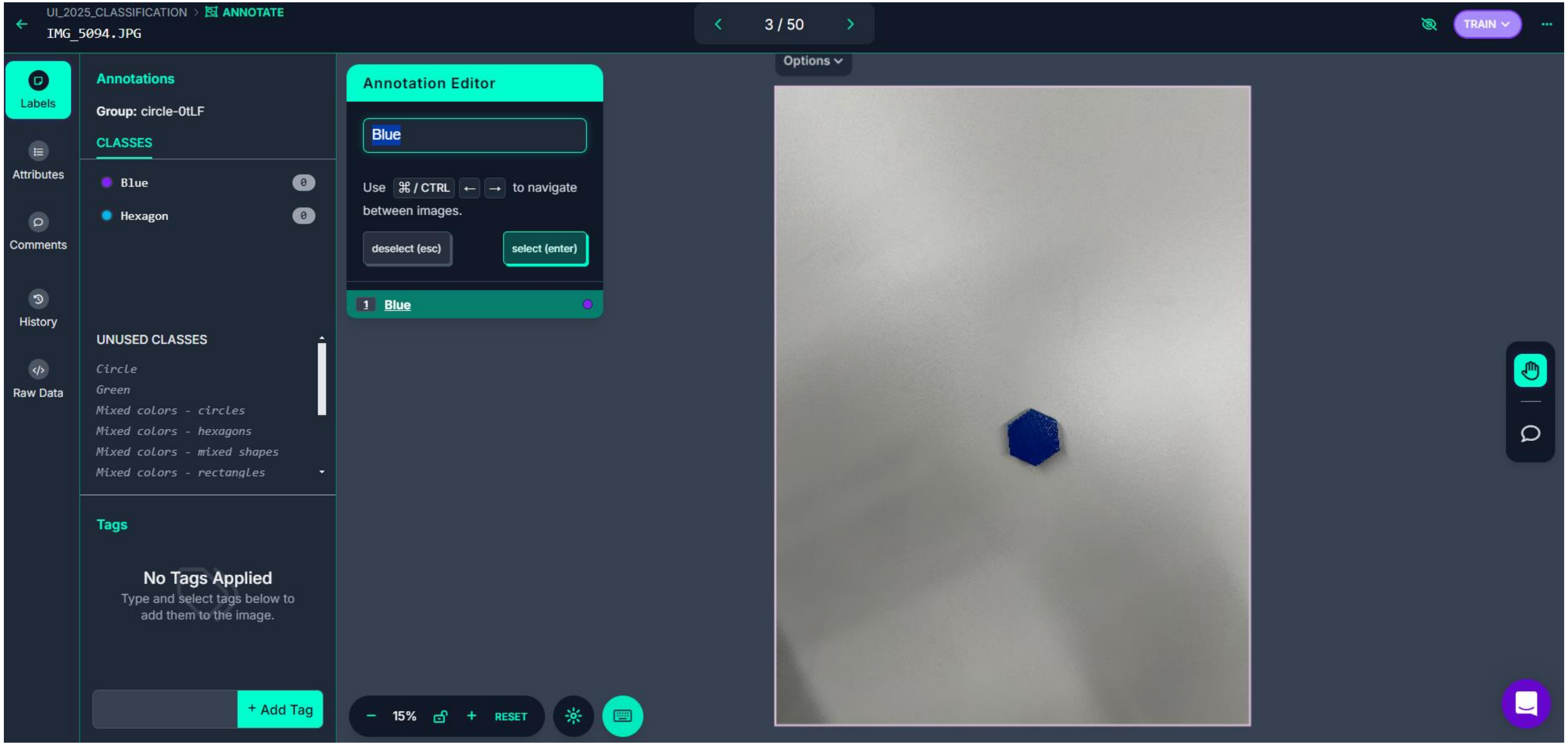


Zadatak 3: Instance classification

- Cilj: Napraviti set podataka (označiti tražene slike prema danim oznakama) za treniranje YOLOv8 modela
- Potrebno je preuzeti set slika (15 slika) i klase (oznake) prema kojima će se slike označavati
- Uploadati slike i oznake u Roboflow i označiti ih prema pravilima:
  - Za instance classification potrebno je odrediti koje se sve klase od ponuđenih nalaze na slici

Dataset Link: UmjetnaInteligencija2025Datasetovi

Annotate - Classify



Zadatak4(Optional): YOLOv8 training

- Trening se provodi u Pythonu korištenjem ultralytics librarija
- Za trening je potrebno odrabati model, specificirati korišteni dataset, specificirati veličinu slike, broj epoha treniranja, te ostale hiperparametre po potrebi kako bi se dobili što bolji rezultati tijekom treniranja modela.

Example

Python CLI

```python
from ultralytics import YOLO

# Load a COCO-pretrained YOLOv8n model
model = YOLO("yolov8n.pt")

# Train the model on the COCO8 example dataset for 100 epochs
results = model.train(data="coco8.yaml", epochs=100, imgsz=640)
```

YOLOv8 training parameters

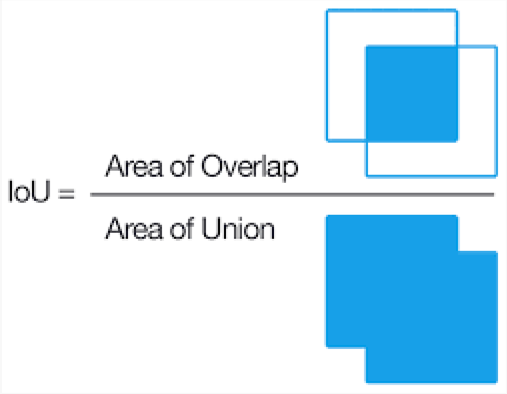
Argument	Type	Default	Description
`model`	`str`	`None`	Specifies the model file for training. Accepts a path to either a `.pt` pretrained model or a `.yaml` configuration file. Essential for defining the model structure or initializing weights.
`data`	`str`	`None`	Path to the dataset configuration file (e.g., `coco8.yaml`). This file contains dataset-specific parameters, including paths to training and validation data, class names, and number of classes.
`epochs`	`int`	`100`	Total number of training epochs. Each epoch represents a full pass over the entire dataset. Adjusting this value can affect training duration and model performance.
`time`	`float`	`None`	Maximum training time in hours. If set, this overrides the `epochs` argument, allowing training to automatically stop after the specified duration. Useful for time-constrained training scenarios.
`patience`	`int`	`100`	Number of epochs to wait without improvement in validation metrics before early stopping the training. Helps prevent overfitting by stopping training when performance plateaus.
`batch`	`int`	`16`	Batch size, with three modes: set as an integer (e.g., `batch=16`), auto mode for 60% GPU memory utilization (`batch=-1`), or auto mode with specified utilization fraction (`batch=0.70`).
`imgsz`	`int` or `list`	`640`	Target image size for training. All images are resized to this dimension before being fed into the model. Affects model accuracy and computational complexity.
`save`	`bool`	`True`	Enables saving of training checkpoints and final model weights. Useful for resuming training or model deployment.

YOLOv8 training

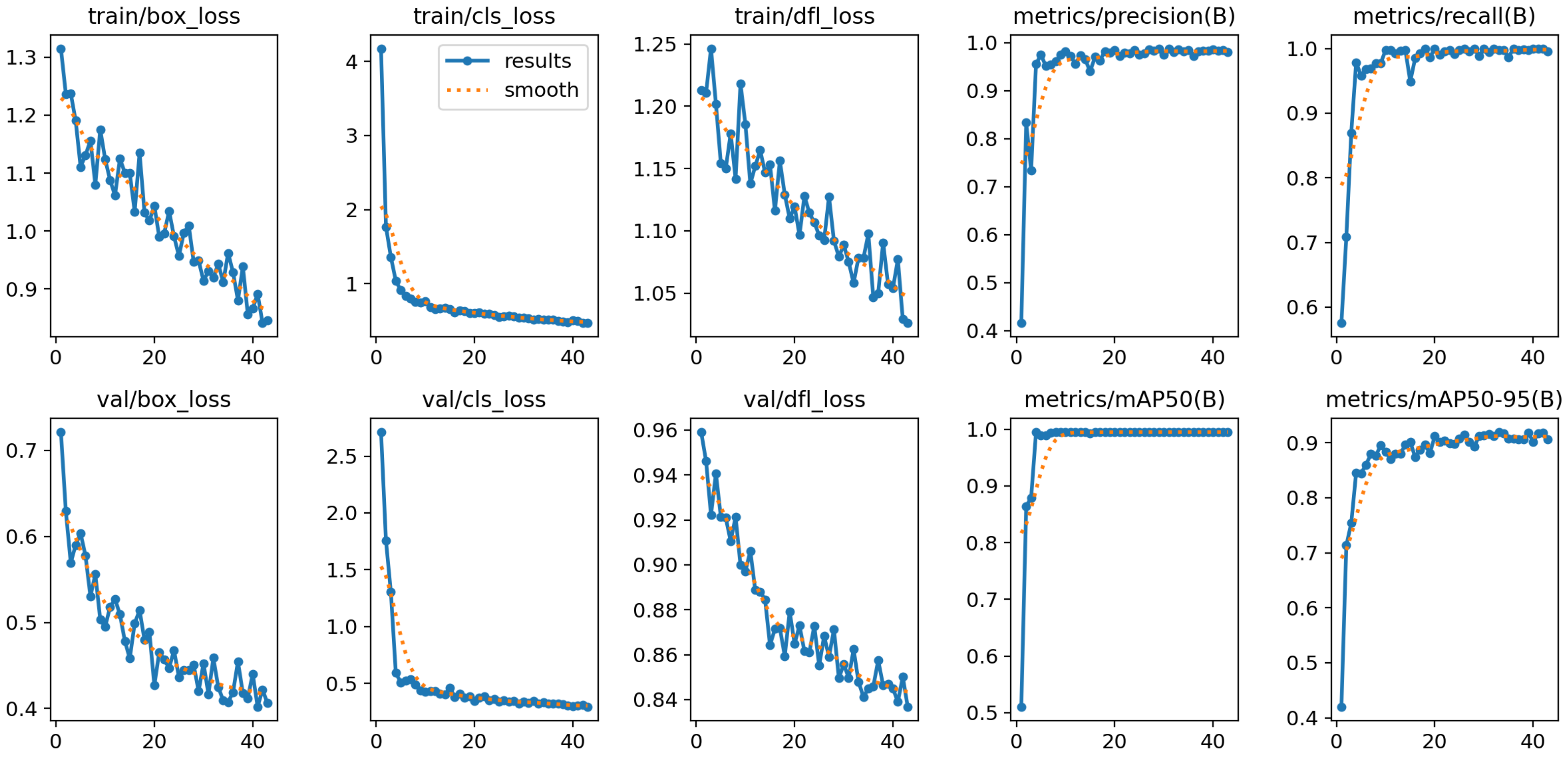
- Za treniranje modela moguće je koristiti Google Colab, alat koji daje pristup Google-ovim računskim jedinicama
- Tijekom treniranja modela za detekciju i segmentaciju prate se sljedeći parametri:
  - Box loss – gubitak koji označava koliko je predviđeni box/poligon treniranog modela različit od stvarnog bounding boxa/poligona
  - Classification loss – gubitak koji označava kolika je netočnost klasifikacije objekata unutar označenog boxa/poligona
  - Cls loss – gubitak koji označava kolika je netočnost detekcije i klasifikacije slabije reprezentiranih klasa unutar seta podataka
  - **Tijekom treniranja je cilj povratnom propagacijom i računanjem gradijenata smanjiti navedene gubitke**

YOLOv8 training

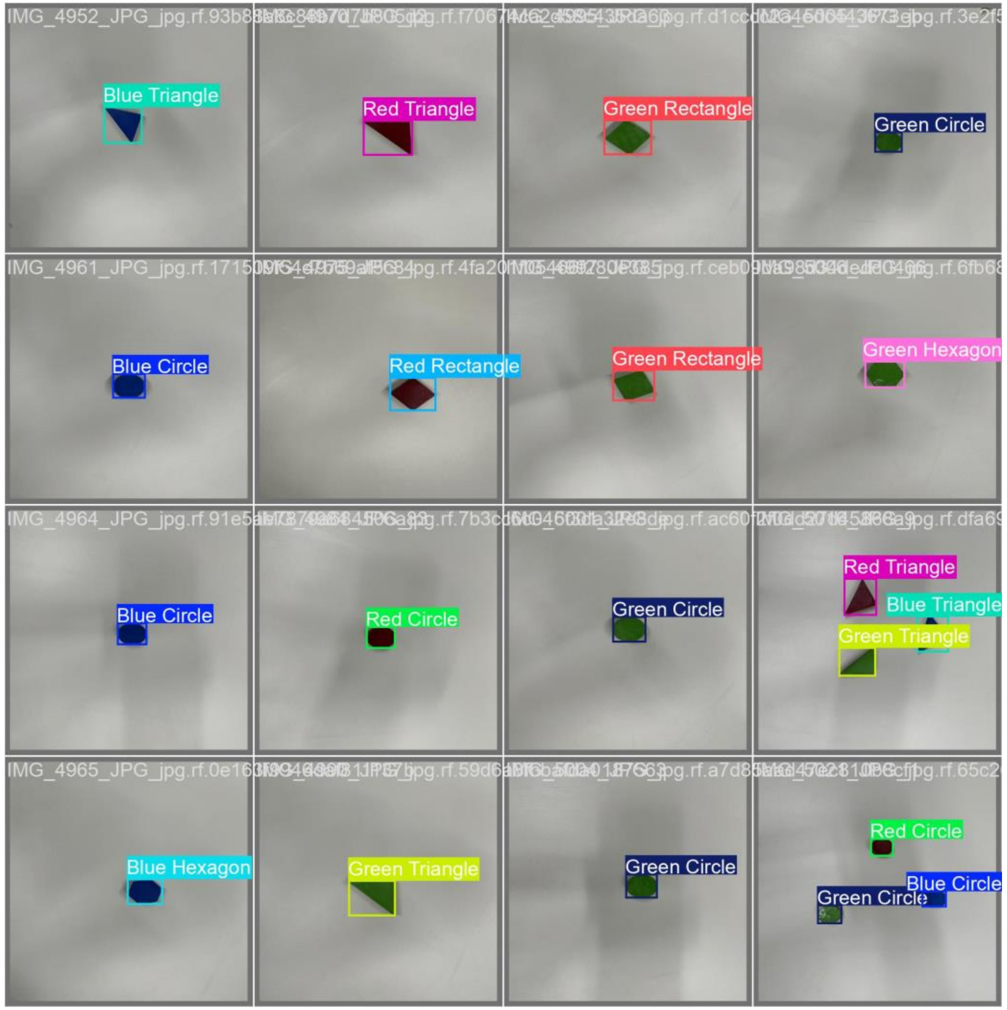
- Osim gubitaka, potrebno je pratiti i točnosti:
  - Precision – omjer točnih predikcija modela (True Positive) u odnosu na sve predikcije modela (True Positive + True Negative)
  - Recall – omjer broja točnih predickija modela (True Positive) u odnosu na sve moguće predikcije koje model može imati (True Positive + False Negative)
  - Mean Average Precision 50 – omjer točnih predikcija modela za Intersection over Union +50%
  - Mean Average Precision 95 - omjer točnih predikcija modela za Intersection over Union +95%



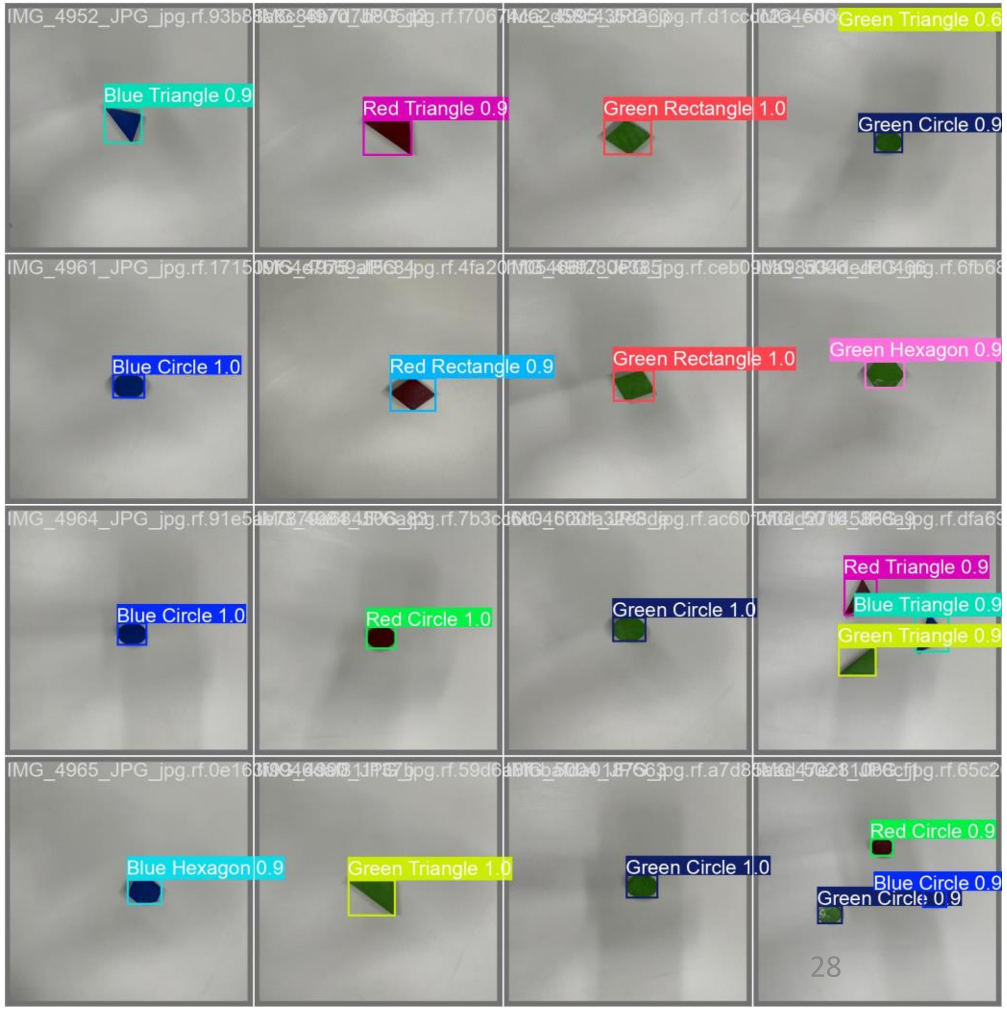
YOLOv8 training rezultati



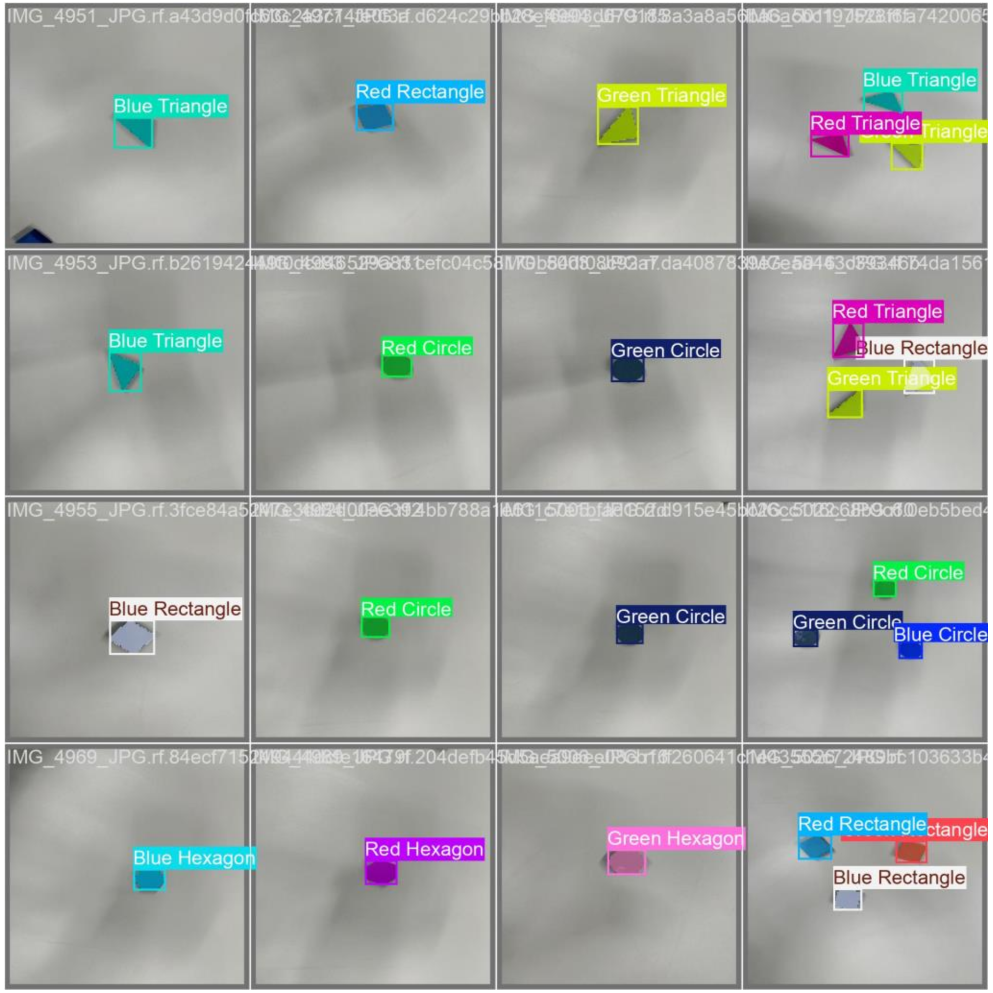
YOLOv8 training rezultati



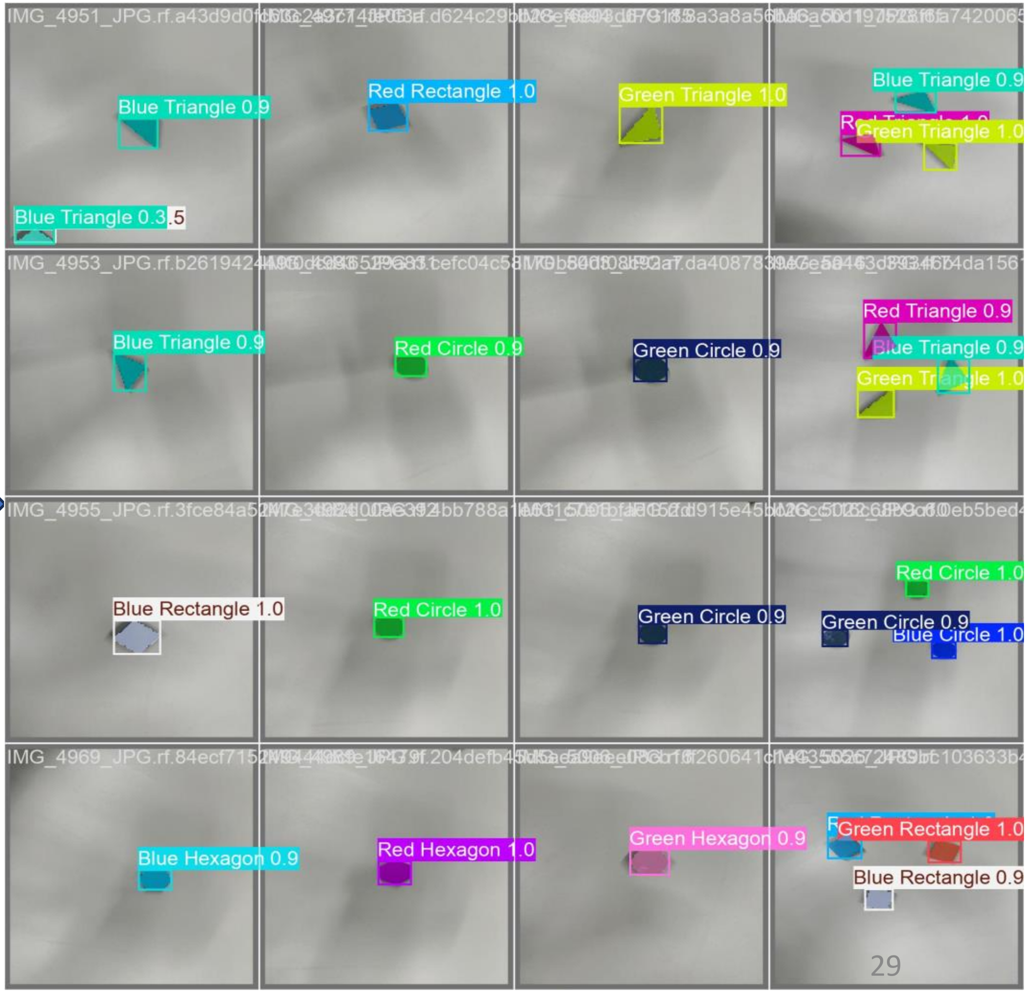
Rezultati



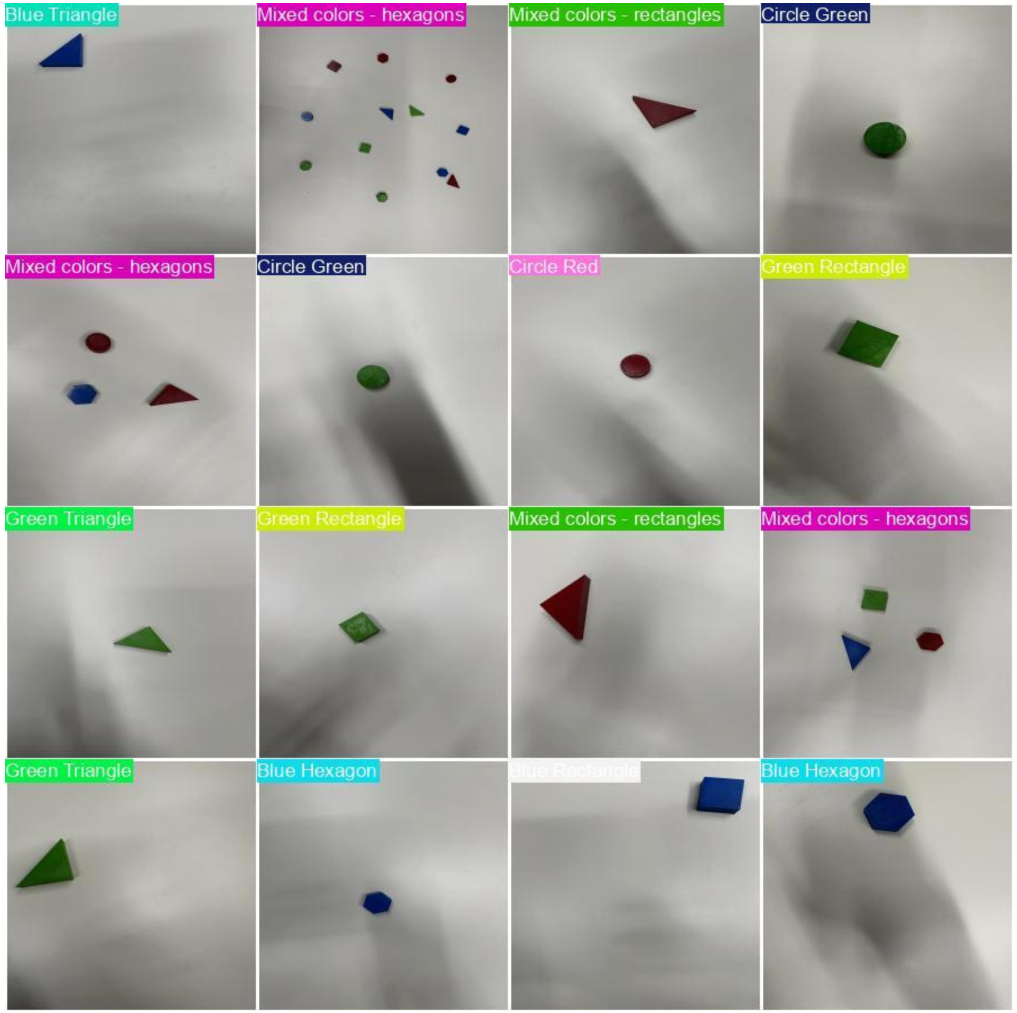
YOLOv8 training rezultati



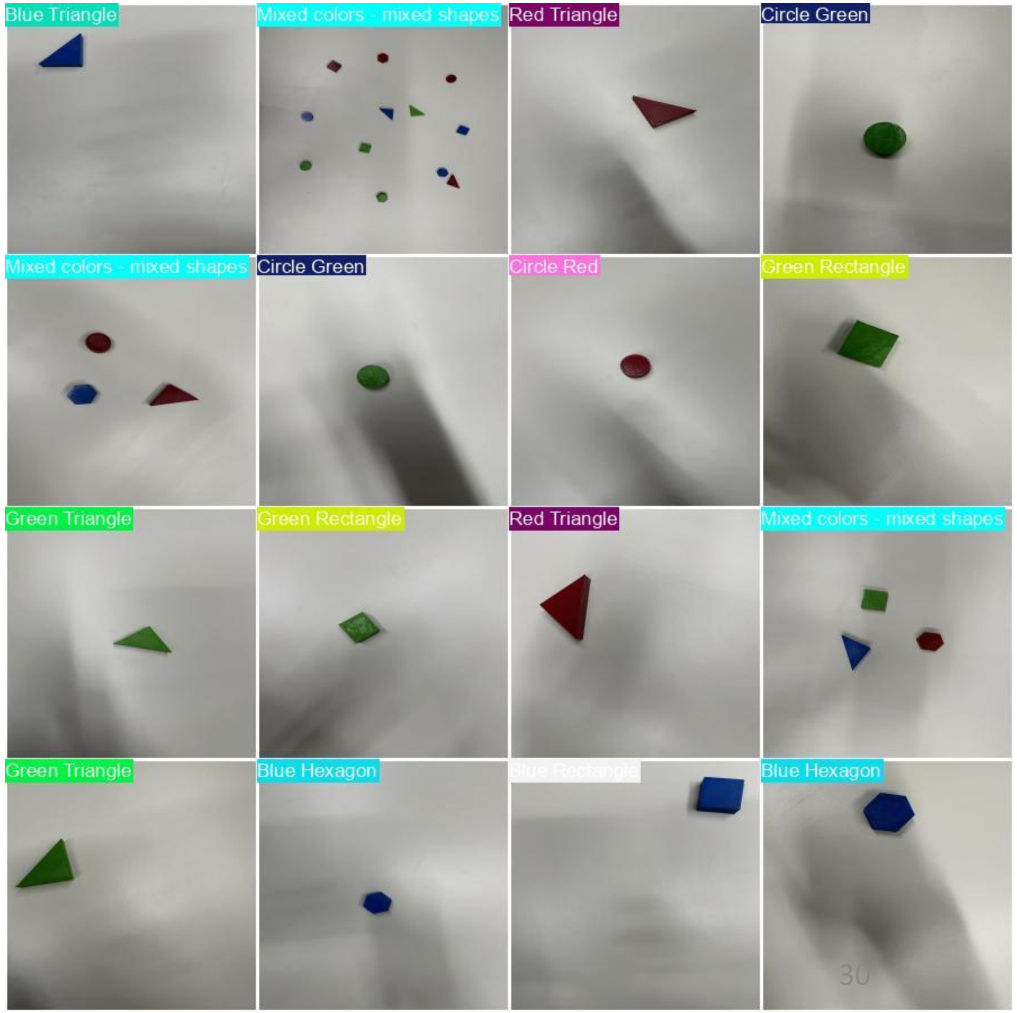
Rezultati



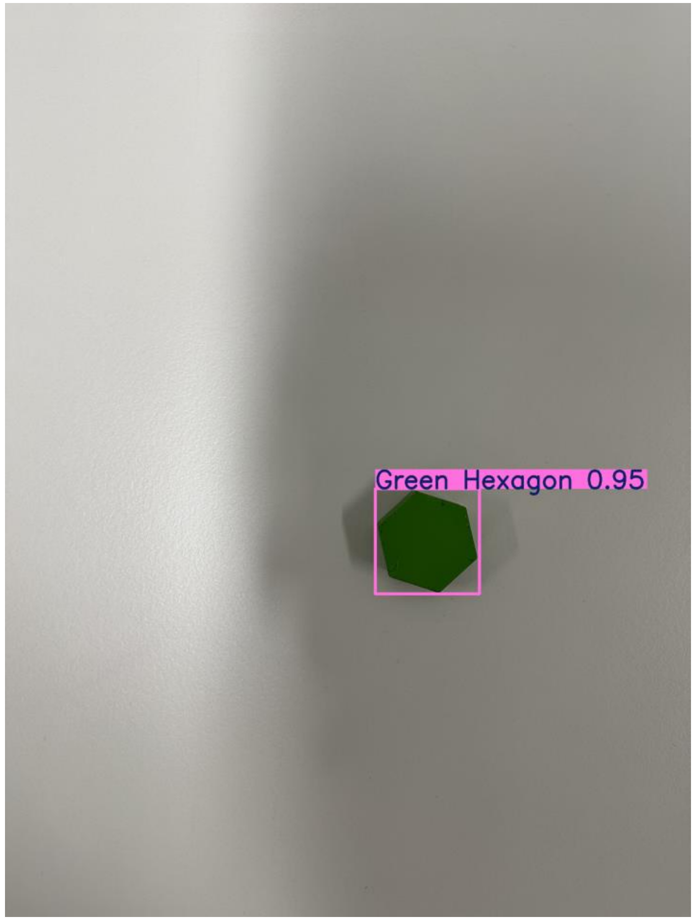
YOLOv8 training rezultati

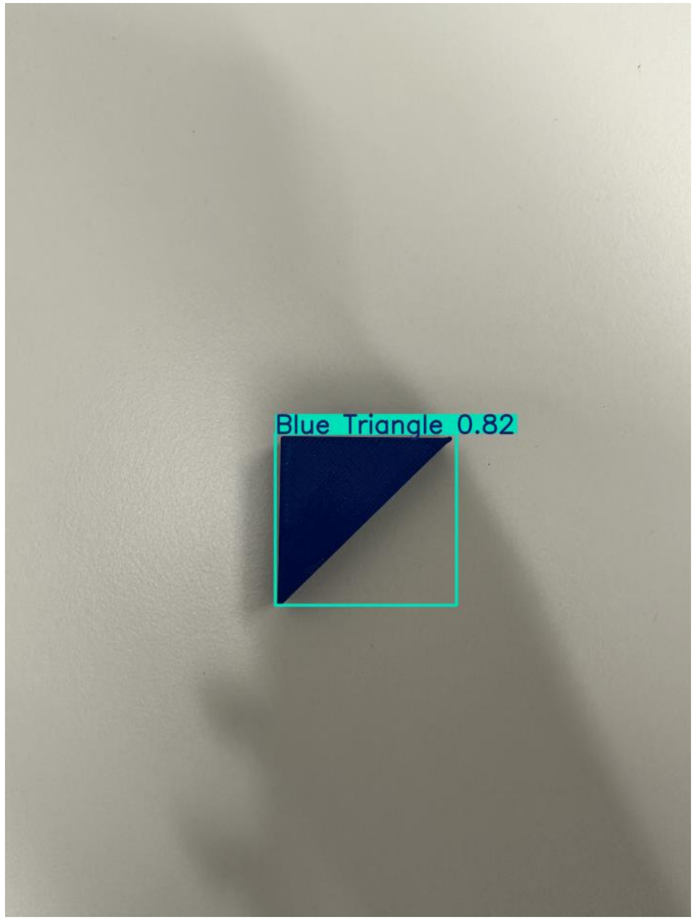


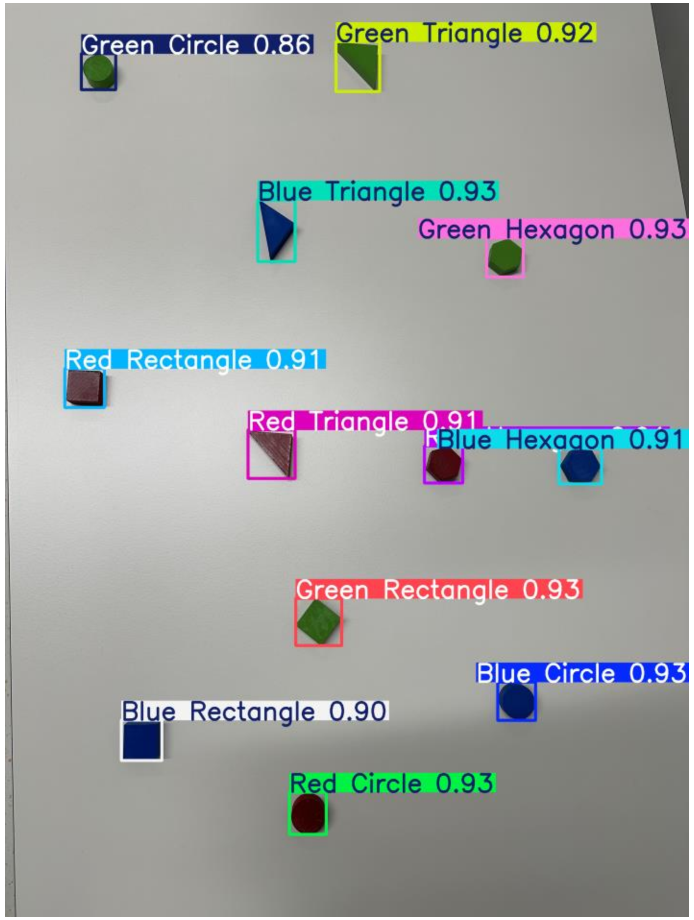
Rezultati



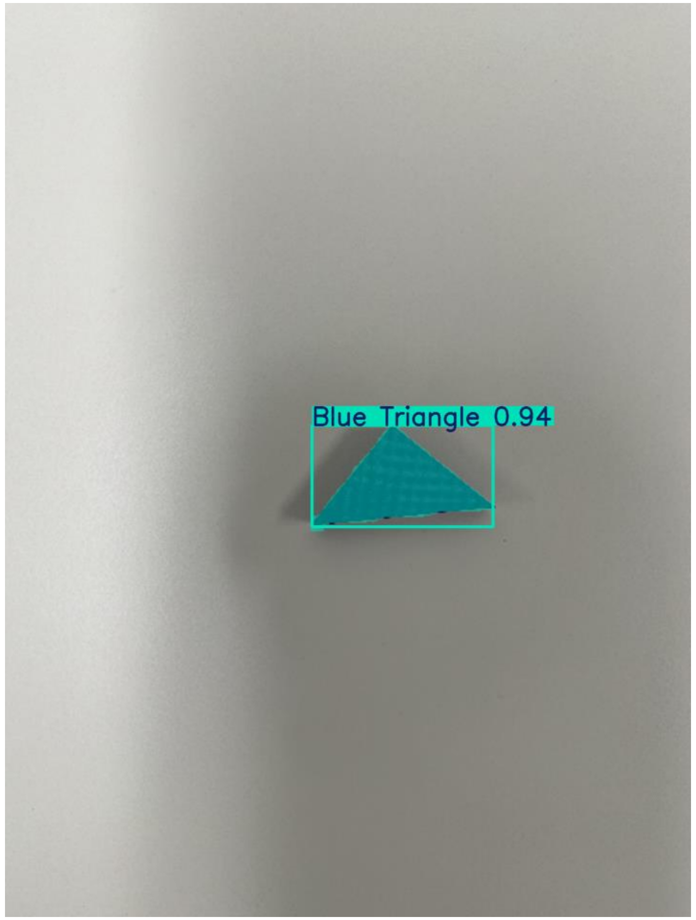
Results – object detection

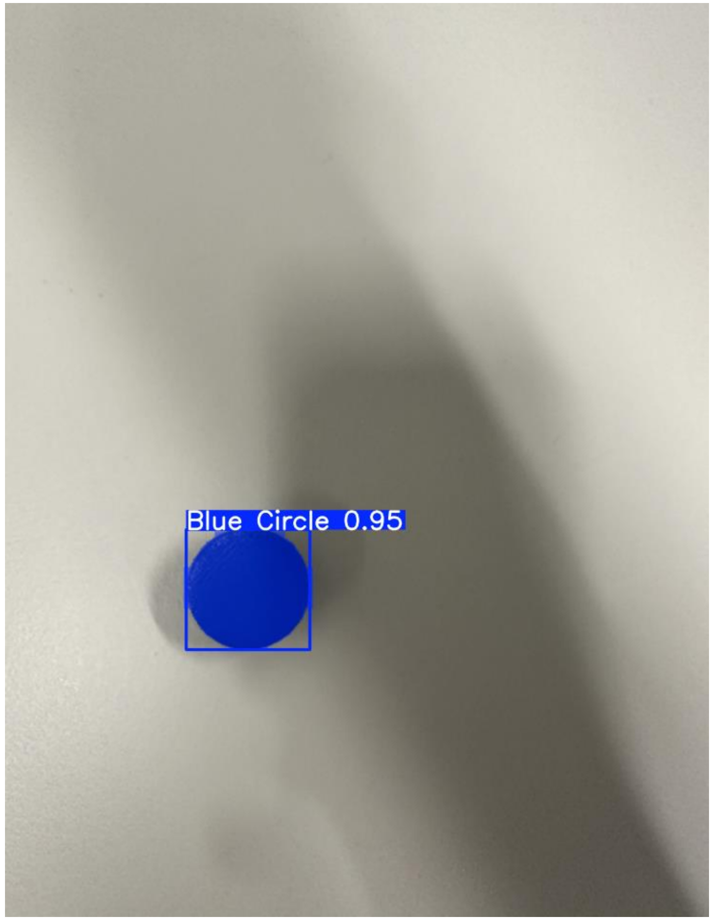


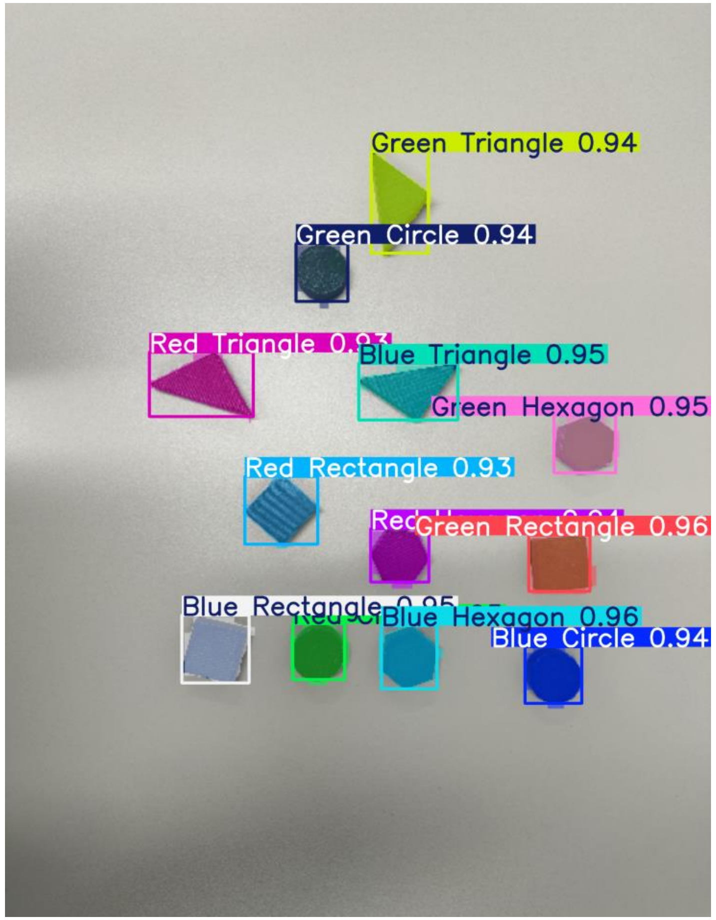




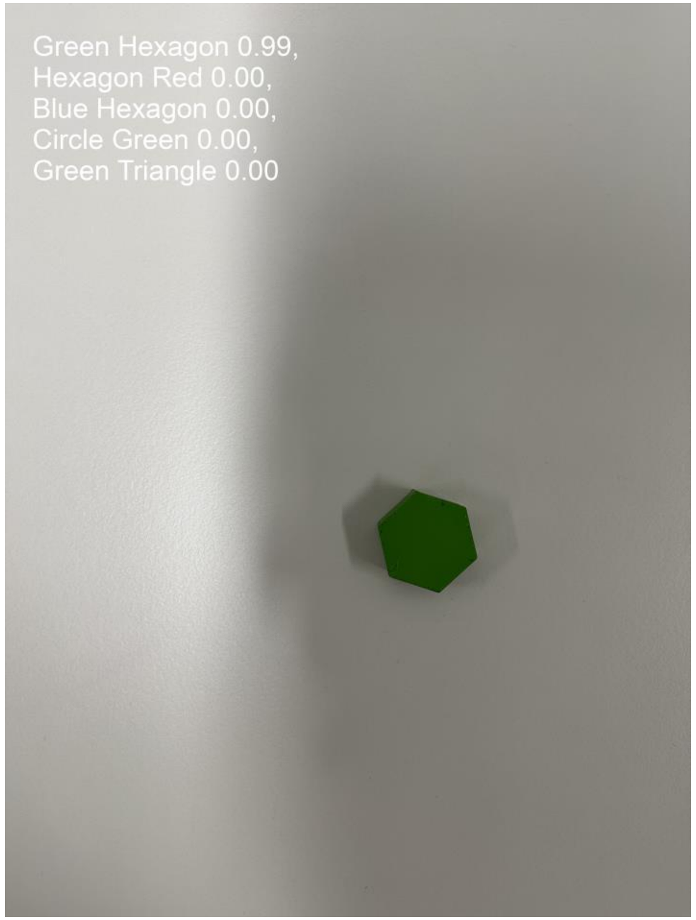
Results – instance segmentation

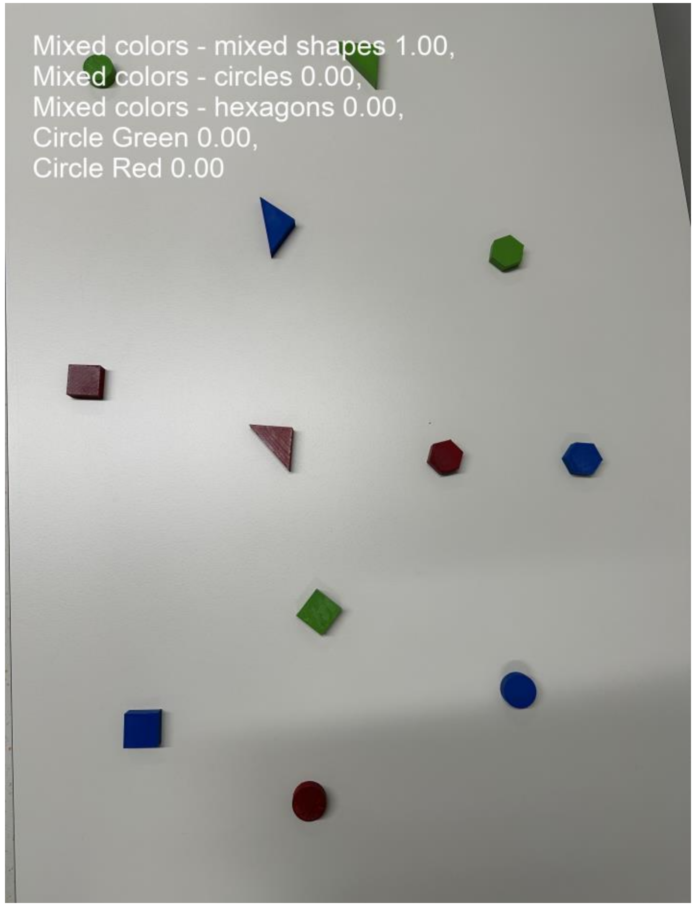






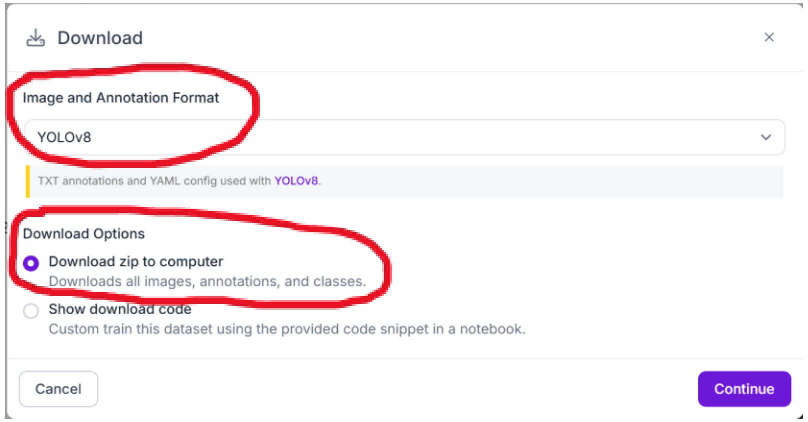
Results - classification

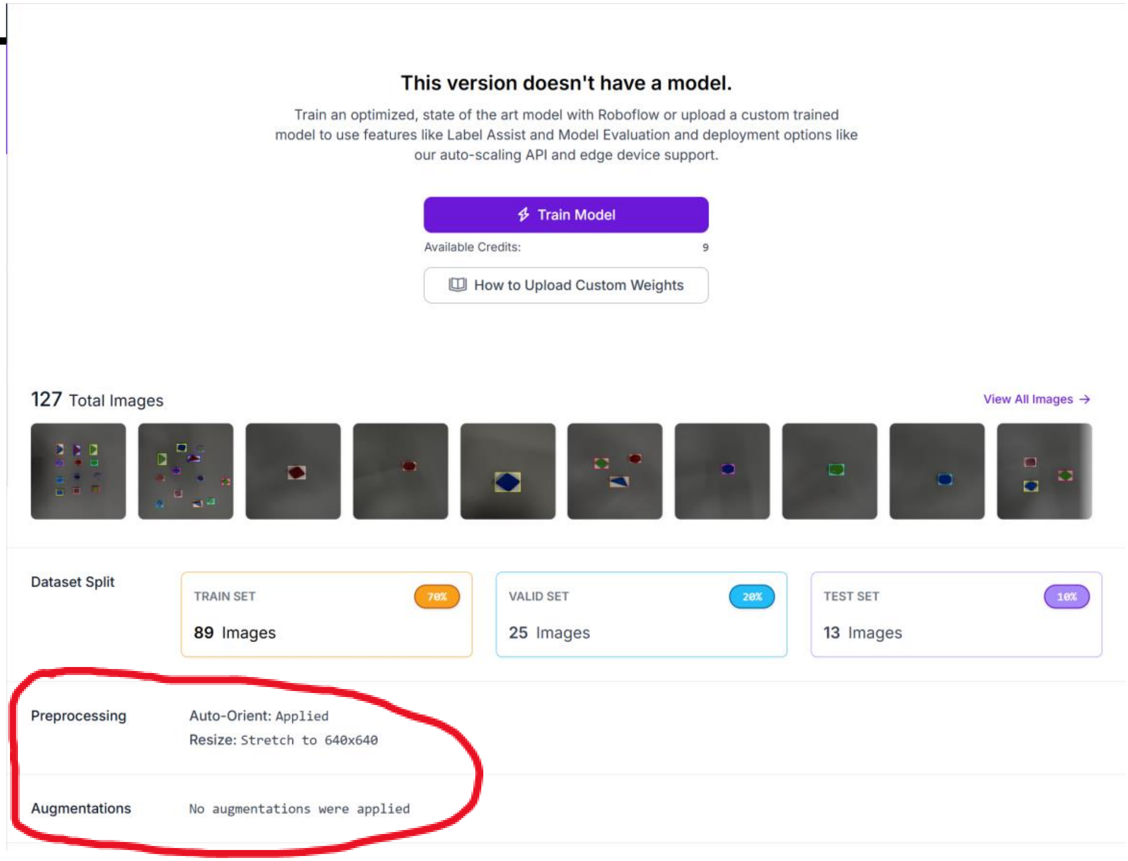




Spajanje datasetova

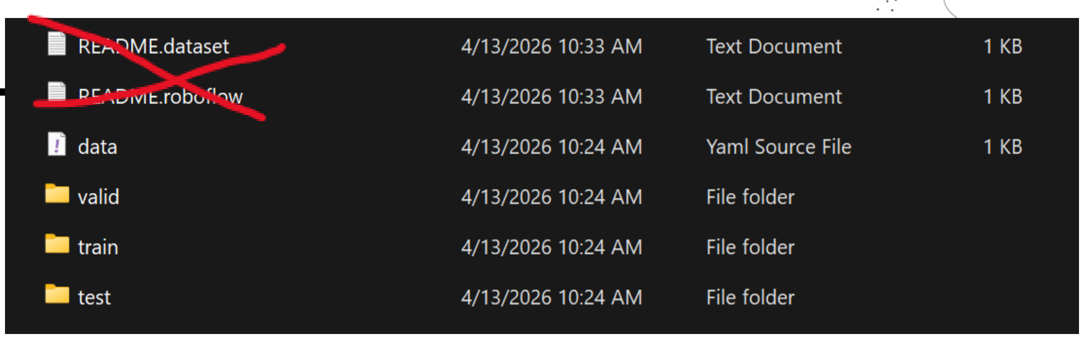
- Stvaranje grupnog dataseta:
  - Nakon što je svatko označio vlasiti dataset potrebno je:
    - Napraviti verziju dataseta bez ikakvih annotation postavki
    - Preuzeti .zip kreiranog dataseta

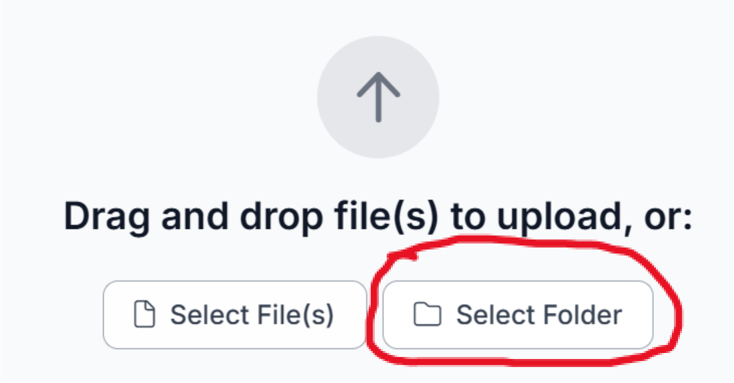




Spajanje datasetova

- Stvaranje grupnog dataseta:
  - Nakon što je svatko označio vlasiti dataset potrebno je:
    - Napraviti verziju dataseta bez ikakvih annotation postavki
    - Preuzeti .zip kreiranog dataseta
    - Raspakirati .zip i obrisati README .txt file-ove
    - Nakon toga u Roboflow-u otvoriti novi projekt te uploadati raspakirani folder (Select Folder) sa svim ostalim folderima unutar
    - Pogledati jesu li anotacije učitane na slikama te dodati slike u dataset





Uploadanje klasa/labela

Potrebno je uploadati klase/labele na početku procesa.
Klase/labeli su spremljeni u **.csv* datoteku.